AI POWERED ML BASES E wardrobe

R26 - IT - 053

Project Proposal Report

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MARCH 2026

AI-Based Outfit Visualization with Animated Fitting

Project ID – R26-IT-053

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DECLARATION

I declare that this proposal report is my own original work, conducted under the supervision of academic staff at the Department of Information Technology, Sri Lanka Institute of Information Technology. This work has not been submitted previously for any degree or diploma at any other institution.

The research presented in this proposal aims to develop an AI-driven outfit recommendation and matchmaking system that integrates semantic knowledge graphs, deep learning-based feature extraction, and diffusion models for generative outfit recommendations. All sources used in the preparation of this proposal have been duly acknowledged through proper citations and references.

I confirm that I have read and understood the academic integrity policies of Sri Lanka Institute of Information Technology and agree to abide by them throughout the duration of this research project.

_____

Student Signature

Date: _____

_____

Supervisor Signature

Date: _____

Abstract

The rapid advancement of digital technologies has revolutionized numerous industries, including fashion and online retail. Users continue to struggle when it comes to finding outfits that complement their style, body type, and preferences. Traditional wardrobe management solutions do not give users an effective way to see how clothing combinations would look before wearing them. This limitation frequently results in poor wardrobe selection, repetitive clothing choices, and wasteful purchases.

This project presents an AI-Based Outfit Visualization System with Animated Fitting, which enables users to digitally inspect and assess outfit combinations before physically wearing them. The system analyzes clothes photographs stored in a user's digital wardrobe using Artificial Intelligence (AI), Computer Vision, and 3D visualization techniques.

The AI model categorises clothing products and suggests appropriate outfit combinations based on style compatibility and user preferences.

The system also has an animated virtual fitting feature, which shows clothing items on a digital avatar created from the user's uploaded image. The technology uses facial detection, body approximation, and real-time rendering algorithms to allow users to see how different clothing look on a personalized avatar. This helps users make better decisions and improves the entire wardrobe management experience.

The proposed solution aims to promote wiser wardrobe usage, boost outfit selection efficiency, and promote sustainable fashion practices by encouraging consumers to make better use of their present clothing items. To provide an engaging and scalable user experience, the system will be constructed using cutting-edge technologies such as machine learning frameworks, computer vision libraries, and mobile development platforms.

Keywords: Artificial Intelligence (AI), Computer Vision, Outfit Recommendation, Virtual Try-On, Animated Fitting, Digital Wardrobe, 3D Avatar Visualization, Machine Learning, Fashion Technology, Augmented Reality.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
GNN	Graph Neural Network
KG	Knowledge Graph
NLP	Natural Language Processing
XAI	Explainable Artificial Intelligence
ResNet	Residual Network
ReLu	Rectified Linear Unit
API	Application Programming Interface
JSON	JavaScript Object Notation
SQL	Structured Query Language
GPU	Graphics Processing Unit

1. INTRODUCTION

Fast fashion and impulsive clothing purchases cause significant textile waste worldwide. Consumers frequently forget items they already own because wardrobes lack visibility and simple tracking. E-Wardrobe is a mobile-first web application that helps users digitise their wardrobe, automatically categorise items using an in-browser AI classifier (Google MobileNet via TensorFlow.js), and promote sustainable usage through outfit matchmaking and wear tracking.

The global fashion industry has grown rapidly in the last 10 years as a result of fast fashion, which entails producing clothing swiftly and selling it at low rates. This strategy increases people's access to clothing, but it also increases textile waste and usage. Because their closets are disorganized and they are unable to see what they own, many consumers make impulsive purchases of clothing without understanding they already have identical goods. As a result, a large amount of clothing is never worn, creating needless waste and environmental issues.

According to studies, a large percentage of items in personal wardrobes are rarely worn. Consumers frequently feel decision fatigue when selecting ensembles because their clothing pieces are dispersed, poorly organized, or not readily visible. This results in repeated acquisition of comparable goods and underuse of existing clothing. As a result, the fashion sector continues to generate excessive textile waste, putting strain on landfills and natural resources.

With the increasing use of smartphones and low-cost internet connectivity, mobile-first digital solutions have arisen as a viable way to handle daily lifestyle concerns. In the fashion industry, digital wardrobe applications are designed to help consumers manage their clothes collections more efficiently. However, many existing wardrobe systems rely on manual item categorization, which can be time-consuming and cumbersome for users. This reduces the practicality and long-term utility of such systems.

Recent breakthroughs in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision have created new potential to automate clothing management. Image recognition models can examine garment photographs and automatically identify them based on variables like category, color, and style. These technologies enable the creation of intelligent systems that not only digitally organize closets, but also provide tailored outfit recommendations and insights into clothing consumption.

The suggested AI-based, ML-powered E-Wardrobe solution tackles these issues by allowing customers to digitize their wardrobe items with simple smartphone photos. To automatically classify clothing items in the browser, the system uses a trained machine learning model based on MobileNet and TensorFlow.js. This technique enables effective

picture recognition without the need for complex server-side infrastructure, making the system lightweight and suitable for mobile-first scenarios. Following classification, clothing items are placed in a searchable digital wardrobe that allows users to quickly manage, filter, and display their apparel. Furthermore, the system monitors wear frequency by noting how frequently each item is used, allowing users to discover rarely worn apparel and encouraging more mindful and sustainable clothing use.

Aside from basic clothing management, the system incorporates Artificial Intelligence, Augmented Reality (AR), and 3D visualization technologies to improve user interaction. A 3D avatar-based outfit visualization tool lets users explore clothing combinations on a virtual model of their body measurements. Users may see how recommended outfits appear on the avatar before selecting them, which improves decision-making and confidence.

To produce compatible outfit combinations, the outfit matchmaking engine analyses apparel qualities such as category, color, and style. The E-Wardrobe system encourages smarter clothing utilization while reducing unnecessary fashion consumption by combining AI-driven clothing classification, trained machine learning models, digital wardrobe management, AR-enabled 3D avatar visualization, outfit matching, and wear-pattern analytics. This sophisticated and engaging technique enables consumers to uncover underutilized apparel pieces while also encouraging more sustainable fashion behavior.



Figure 1

1.1 **Background**

Choosing appropriate clothing combinations is a typical difficulty for many consumers, particularly when wardrobes contain a big number of pieces that are difficult to imagine together. Users frequently struggle to visualize how different items would look when worn without actually trying them on. This limitation results in wasted time, repetitive outfit choices, and underuse of available clothing pieces. Furthermore, existing digital wardrobe applications primarily display static photographs of apparel, which may not adequately reflect how garments fit or appear on the actual body. Current outfit visualization approaches are limited because they do not accurately represent real-world wearing situations. Users cannot easily assess garment fit, movement, or styling compatibility with mere visual previews. As a result, users continue to rely on human trial and error to select outfits.

Limitations of Current Outfit Visualization Methods

Existing solutions primarily depend on:

• Static garment graphics with no body representation.
• Basic 2D outfit previews.
• Expensive smart mirror systems.• Specialized 3D scanning hardware.

Reasons why AR-Based Avatar Outfit Visualization is Effective

Recent research in computer vision, augmented reality, and virtual fashion technologies suggests that merging 3D avatars with AR-based visualization might successfully imitate clothing appearance and improve outfit decision-making. With advancements in graphics techniques and body-modelling algorithms, virtual try-on systems can now show realistic apparel previews. These technologies enable:

• Realistic outfit visualization on a 3D avatar using body measurements.
• Use animated rendering to simulate clothing fitting and movement.
• A dynamic preview of AI-recommended clothing combinations.
• Improved the user's understanding of style compatibility and clothes appearance.

Modern smartphones and web browsers offer AR frameworks and real-time 3D rendering, making this approach more accessible. As a result, AR-based avatar representation offers a scalable, interactive, and cost-effective alternative for enhancing digital wardrobe experiences and assisting users in making better clothing decisions.

The E-Wardrobe system includes the AI-Based Outfit Visualization with Animated Fitting module as follows:

• The virtual fitting engine, which enables users to preview clothing on a personalized 3D avatar.
• An augmented reality visualization module that dynamically presents recommended outfits with realistic animations and clothing movement.
• The outfit interaction interface allows users to experiment with and assess garment

1.2 Literature Survey / Review

1.2.1. 3D avatar creation system

Artificial intelligence (AI), computer vision (CV), augmented reality (AR), and 3D modeling technologies have improved greatly in recent years, allowing for the creation of interactive systems such as virtual try-on apps and AI-powered wardrobe management systems. These technologies enable users to perceive clothing goods on digital avatars or in augmented reality, boosting personalization and decision-making in the fashion and e-commerce industries. This section examines existing approaches, including methodology, performance, and limits.

Early virtual try-on systems used simple image overlay techniques to show clothing or accessories on a user's photo. These technologies lacked accurate body modelling and realistic movement simulation, resulting in a limited user experience. As deep learning and computer vision have advanced, new virtual try-on systems now include 3D reconstruction, position estimation, and real-time rendering.

Marelli et al. [1] suggested an artificial intelligence-based virtual try-on online tool that uses deep learning to create a 3D face model from a single image. The device enables users to see eyewear on their recreated 3D face from various perspectives. The system combines facial detection, position estimation, and 3D rendering modules to provide a realistic try-on experience. While the technology improved user interactivity and accessibility by using a web-based interface, it was limited to visualizing facial accessories like spectacles rather than full-body apparel.

Recent research has looked into 3D avatar synthesis from single photographs to increase the realism of virtual try-on systems. High-quality synthetic persons have been created using generative AI models such as DreamGaussian and Kolors, which preserve facial texture and identity. These systems can create entire avatars from a single frontal photograph and place them in real-time interactive surroundings. Experimental results show segmentation accuracy above 0.92 Intersection-over-Union (IoU) and real-time rendering speeds of about 64 frames per second, allowing for fluid interaction in virtual worlds.

Real-time motion synchronization has also been investigated using pose estimation techniques such as BlazePose, which tracks body movements with a single RGB camera. A user's movements can be linked with a digital avatar in real time by combining posture estimation and inverse kinematics. The System Usability Scale (SUS) was used in usability studies, and scores of roughly 92.4/100 were obtained, showing great usability and efficacy for interactive applications including virtual fitting rooms and digital twin settings. In comparison to classic marker-based motion capture (MoCap) systems, AI-based pose estimate algorithms considerably reduce hardware requirements and costs. Marker-based systems often necessitate infrared cameras and wearable markers, making them costly and difficult to use in consumer applications. Computer vision-based pose tracking, on the other hand, may be used with common

RGB cameras, making it ideal for scalable and mobile virtual try-on systems. Several commercial applications showcase the practical application of AR-based virtual try-on technology. YouCam Makeup, Jeeliz, and Ditto Virtual Try-On are three platforms that allow users to visualize accessories and cosmetics in real time using augmented reality. However, many of these systems use 2D overlays instead of complete 3D avatar-based fitting, which diminishes realism and accuracy when the user changes stance or viewing angle. Modern virtual try-on systems often follow a pipeline that includes user identification, position tracking, avatar generation, apparel rendering, and scene mixing modules. These elements work together to create an interactive augmented reality scene in which digital apparel is realistically aligned with the user's body. However, difficulties persist in areas such as precise body shape estimation, clothing deformation simulation, and realistic animation during complex movements.

Despite these limits, the integration of AI, augmented reality, and 3D avatar technology has substantially enhanced the possibilities of virtual try-on systems, allowing for more immersive and personalized digital fashion experiences.

1.3 Research Gap

Although virtual try-on technology has improved substantially, there are still various constraints. Many existing systems focus on accessories such as eyeglasses or makeup rather than providing a comprehensive garment representation with dynamic body movement. Some systems require many cameras or specialized motion capture technology, increasing costs and limiting accessibility.

Furthermore, many virtual try-on apps employ static 2D overlays or pre-recorded animations, which do not imitate actual clothing motion while the user walks. Accurate synchronization of human movement and garment animation remains difficult, particularly when only one RGB camera is utilized.

To solve these limitations, the suggested solution employs augmented reality and 3D avatars to provide AI-based outfit visualisation with animated fitting. The system's goal is to provide real-time motion synchronization, realistic outfit visualization, and interactive outfit preview, allowing users to explore clothing combinations dynamically while requiring little technology and ensuring high accessibility.

Table: Comparison of Existing Virtual Try-On Systems and Proposed Smart E-Wardrobe System

Feature / Component	[1] AR Try-On Apps	[2] 2D Virtual Fitting	[3] Motion Capture Systems	[4] AI Fashion Apps	[5] Smart E-Wardrobe (Proposed)
Real-time virtual try-on	✓	✓	✓	×	✓
Low-cost deployment (mobile based)	✓	✓	✓	×	✓
Works with single RGB camera	✓	✓	✓	×	✓
Full garment visualization	×	✓	✓	×	✓
Dynamic body movement simulation	×	×	✓	×	✓
3D avatar generaton	×	×	✓	×	✓
AI outfit recommendation	×	×	✓	×	✓
AR based outfit preview	✓	×	✓	×	✓
Face based avatar personalization	×	×	✓	×	✓
Moblle accessibility	✓	✓	×	×	✓

Notes (short):

[1] AR-based virtual try-on aplications [2] 2D clothing overlay systems [4] AI fashion recommendation apps

[2] 2D clothing overlay systems [4] Motion capture based fitting systems [5] Smart mirror AR fitting systems

[**Proposed**: Smart E-Wardrobe AR Visualization System

Table 1 : Research Gap table.

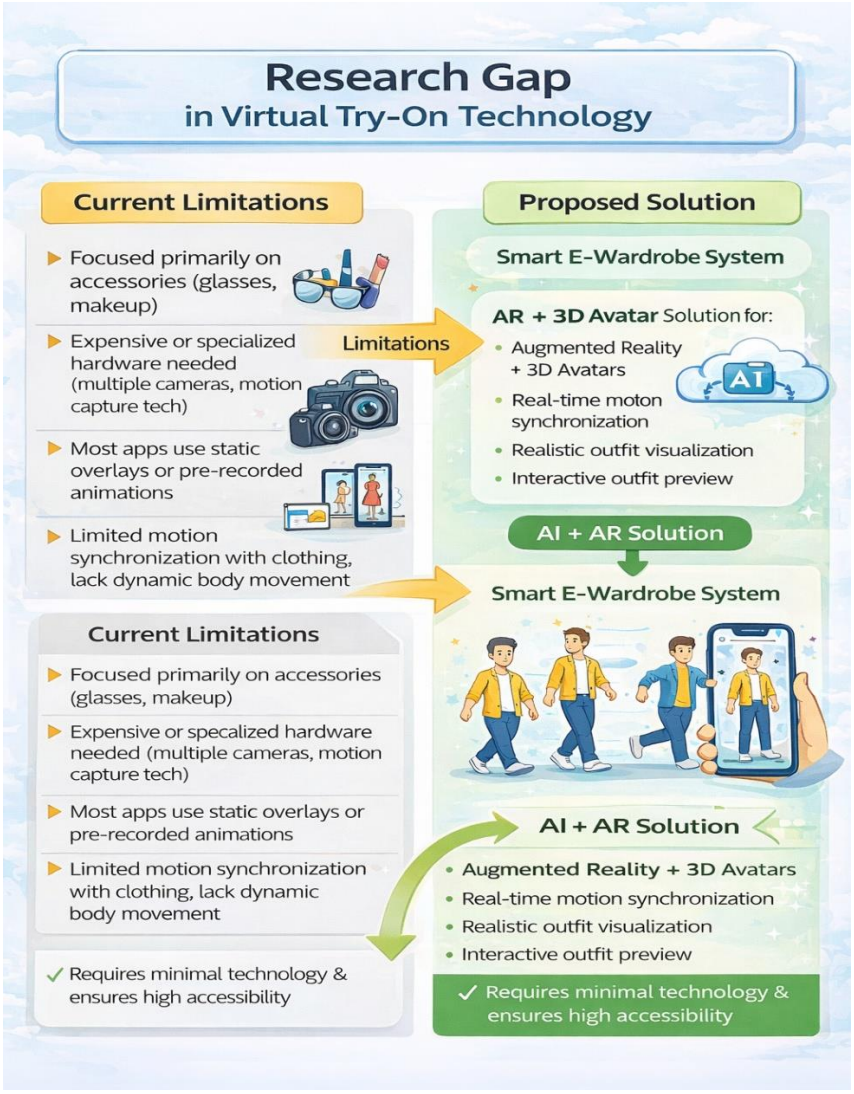


Figure 2 : Research Gap Diagram

1.4 Research Problem

Despite the growing popularity of digital wardrobe applications and fashion recommendation systems, many current solutions are severely limited in terms of automation, usability, and sustainability. Most current wardrobe management apps rely largely on manual data entry and tagging, which is time-consuming and reduces long-term user interest. As a result, users frequently neglect to keep an accurate digital record of their apparel.Many clothing items go unused owing to inadequate wardrobe presentation, clutter, or customers forgetting what they own. This results in wasted garment purchases and contributes to rising textile waste, a significant environmental concern in the worldwide fashion industry (Yalcin-Enis, I., Kucukali-Ozturk, M., & Sezgin, H.).

The fundamental problems of contemporary digital wardrobe systems may be summarized as:

- Manual tagging and organization: Most systems require users to manually upload, crop, and categorize clothing items, which takes time and decreases usefulness.
    - Limited automation: Existing wardrobe programs lack AI-based clothes classification, rendering wardrobe management inefficient and challenging for novices.
    - Lack of realistic outfit visualization: Many platforms simply show static outfit choices, preventing users from visualizing apparel on a 3D avatar or using augmented reality.
    - Inadequate sustainability insights: Current systems rarely measure wear frequency, apparel usage patterns, or sustainability metrics, which are critical for promoting responsible fashion consumption.
    - Low user involvement: Few platforms allow for social sharing, outfit inspiration, or community wardrobe features, limiting user interaction and long-term engagement.

2. OBJECTIVES

2.**1 Main Objective-**

The major goal of this project is to create an AI-powered smart wardrobe and outfit visualisation system that automatically identifies clothing items, organises them into a digital wardrobe, and allows users to visualise ensembles using Augmented Reality (AR) and animated 3D avatars.

The system combines artificial intelligence, computer vision, and augmented reality technologies to increase closet visibility, aid in outfit decision-making, and promote sustainable clothing usage by measuring wear frequency and encouraging the reuse of underutilised articles.

• To develop an interactive and intelligent wardrobe management software that decreases clothing waste while increasing user engagement through automatic garment detection, outfit development, and virtual try-on visualisation.

2.2 Specific Objectives

• To investigate existing wardrobe management applications and textile waste research in order to identify limitations in current systems and determine the essential features required for an intelligent wardrobe solution.

• To design and develop a mobile-first user interface using the MERN stack with React, Tailwind CSS, and shadcnUI, enabling users to easily upload and manage clothing items within a digital wardrobe.

• To implement a computer vision–based clothing classification system using a pre-trained deep learning model such as MobileNet integrated with TensorFlow.js, enabling automatic detection and tagging of clothing categories from user-uploaded images.

• Create a digital wardrobe management system that enables users to store, search, filter, and organise clothing items while retaining item metadata and usage logs.
• Create an AI-driven outfit recommendation engine that suggests daily outfit combinations based on available closet items, categories, and usage trends.
• Implement a wear-tracking and analytics module that tracks how frequently clothing items are worn and highlights the least-worn articles, encouraging greater clothing utilisation and sustainable purchasing behaviour.
• Create an Augmented Reality (AR)-based outfit visualisation tool that enables users to sample outfits on a virtual 3D avatar, allowing for animated fitting and realistic outfit depiction.

Use 3D avatar modelling and animation techniques to imitate garment appearance and body interaction, hence increasing the realism of virtual try-on experiences.
• Evaluate the system's performance and usability using criteria such as categorisation accuracy, user happiness, system responsiveness, and usability testing with potential users.

3. Methodology

3.1 Component Overall System Description

The proposed system is an AI-powered digital wardrobe and virtual outfit visualisation platform that allows users to organise clothing items, create outfit combinations, and sample outfits utilising Augmented Reality (AR) and animated 3D avatars.

Unlike existing wardrobe management tools, which rely significantly on manual labelling and organising, the suggested solution uses AI, computer vision, and augmented reality technologies to automate garment classification, digital wardrobe management, and outfit visualisation. Users can input wardrobe photographs through a mobile or web interface, and the system will automatically identify clothing categories using a machine learning algorithm.

In addition to wardrobe organising, the system includes an AI-powered outfit matchmaking module that creates outfit combinations from available clothing pieces. Users can then upload a photo of their face, which is then processed with face detection techniques to create a personalised 3D avatar. The chosen clothing are applied to this avatar, and animations mimic natural body movements for a more realistic fitting experience. Finally, an augmented reality interface is used to provide the outfit visualisation, allowing viewers to interact with clothing combinations.

The system design is organised into four primary layers that work together to provide intelligent wardrobe management and a virtual try-on environment.

3.2 Data Collection Layer (Wardrobe Image Upload + User Data)

This step allows users to contribute garment photographs as well as personal information required for outfit visualisation.

The system gathers the following data:

- Wardrobe Images - Photos of clothing items taken with smartphones or uploaded directly from the user's device.
- User Profile Data - This includes basic information like gender, clothing preferences, and style classifications.
- Upload a facial image to personalise your virtual avatar.
- Optional Body Measurements - Height or body proportions that aid in scaling the 3D avatar for greater fitting accuracy.

This stage guarantees that the system has enough data to do clothing classification, wardrobe management, and avatar-based outfit visualisation.

3.3 Data Processing and Preparation Layer

This stage prepares the uploaded data for machine learning and visualisation operations.

Important preprocessing tasks include:
- Image preprocessing steps include scaling, normalisation, and background modification.
- Using picture feature extraction to prepare clothing photos for classification.
- Face detection preprocessing to find facial landmarks in uploaded photos.
- Data structure and storage in the digital wardrobe database to facilitate retrieval.

These processes increase the quality of the input data and the performance of the system's AI models.

3.4 Core AI Processing Layer (Clothing Classification + Outfit AI + Avatar Generation)

This layer comprises the system's main intelligence and incorporates a number of AI and computer vision components.

The major modules include:

• Clothing classification (Machine learning)
A pre-trained MobileNet model, implemented in TensorFlow.js, is used to automatically classify clothing items from uploaded photos into categories such as shirts, slacks, dresses, and coats.

• Smart Wardrobe Organization Classified clothing items are organised in a digital wardrobe for easy search, filtering, and management of collections.

• AI-Based Outfit Matchmaking A recommendation module creates outfit combinations based on clothing categories, style compatibility, and wardrobe components.

• Face detection (Computer Vision)
The submitted user image is processed using Python-based face identification algorithms to determine facial landmarks for avatar development.

3.5 • Create a 3D avatar.

A virtual 3D avatar is created based on the identified facial structure and optional body dimensions.

• Face Texture Mapping Avatar models are personalised by mapping the user's facial picture onto them.

• Body Measurement Scaling User-provided measurements are used to alter the avatar's proportions for more realistic garment fitting.

• Apply clothing items from the wardrobe to the 3D avatar model.

3.6 • Animation Simulation

To depict realistic garment behaviour, animation techniques replicate natural body movements such as standing, turning, and walking.

4 AR Visualization and System Deployment Layer

The final stage presents the system's results to the user via an interactive interface.

The system provides:

• Augmented Reality Outfit Visualisation - Users can preview outfits on their 3D avatars via smartphone AR.
• Interactive Avatar Control - Users can rotate the avatar and change attire to test different clothing combinations.
• Mobile and Web Interface - The platform is built as a mobile-first MERN web app with React, Node.js, Express, and MongoDB.
• Cloud Image Storage - Clothing images are kept in cloud services like Cloudinary.

• Real-Time AI Processing - To ensure efficient performance, the AI classification model is performed directly in the browser using TensorFlow.js.

This architecture keeps the system lightweight, scalable, and accessible for daily users, allowing them to quickly manage their wardrobes while visualising ensembles using AI-powered AR technologies.

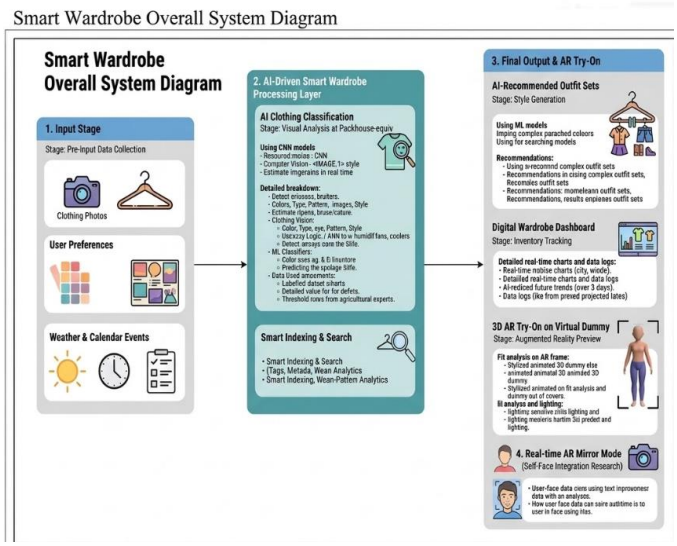


Figure 3 : Overall System Diagram

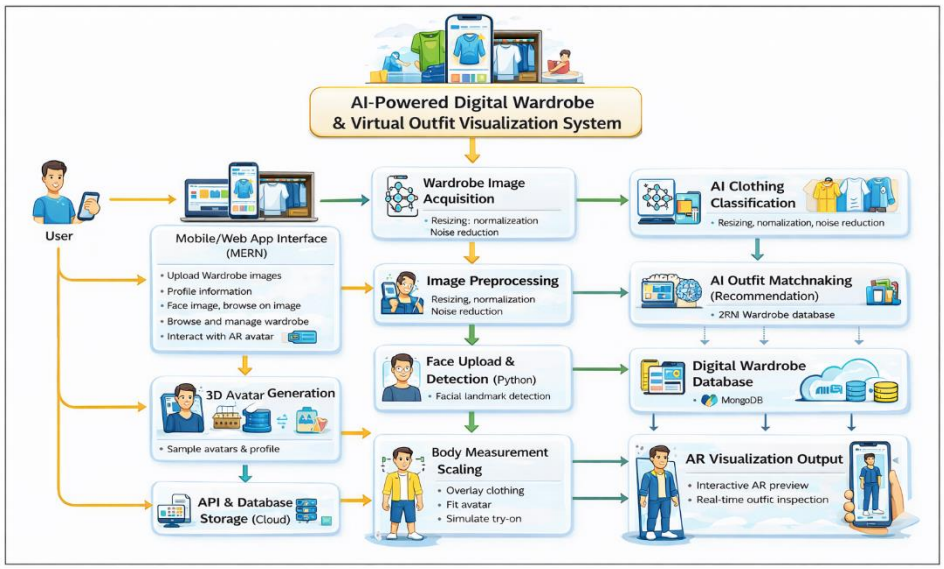


Figure 1: Component Diagram (Component

Figure 4 : component Diagram

4. project Requirements

4.1 Functional Requirements

Functional requirements specify the major processes and services that the system must provide in order to achieve the AI-Based Smart E-Wardrobe with AR Outfit Visualisation goals. These requirements ensure that the system can collect wardrobe photographs, use machine learning to categorise clothing items, generate outfit recommendations, construct a personalised 3D avatar, and visualise clothes with augmented reality.
The system needs to support the following functions:
• Allow users to submit a facial image to personalise their 3D avatar.
• Face Detection and Processing: Use computer vision techniques like OpenCV or MediaPipe to extract the facial region from uploaded images.
• Create a face texture image to customise the 3D avatar model.

• Create a 3D avatar model based on the user's bodily structure.
• Avatar Personalization: Use the retrieved face texture to alter the avatar model based on user body characteristics, including height and proportions.
• Load clothing models based on selected outfit items from the wardrobe.
• Apply clothing models to the 3D avatar to simulate how they will look on the user.
• Simulate avatar actions, including standing, spinning, and walking, to provide realistic clothing behaviour.
• Augmented Reality Visualization: Present animated avatar and outfit in an AR environment for interactive preview.

4.2 Non-Functional Requirements

Non-functional requirements explain the proposed system's quality attributes and operational standards, not the precise functionalities it performs. These requirements ensure that the AI-powered smart wardrobe and AR outfit visualisation system is efficient, dependable, and safe across multiple devices and environments.
These features influence how well the system functions when users input clothing photographs, create avatars, and view ensembles in augmented reality.

• User-Friendliness: The mobile app interface should be easy and straightforward, allowing

users to simply upload wardrobe photographs, create outfits, and visualise apparel on a 3D avatar without technical understanding.

Reliability: The system should consistently perform clothing categorisation, face identification, and avatar rendering without failure. It should still work properly even if user photographs have different lighting, backgrounds, or camera quality.
• Scalability: The system should accommodate more users and wardrobe items without compromising performance. The backend design, which uses Node.js and MongoDB, should be scalable enough to handle big datasets and several simultaneous queries.
• Performance: To provide a responsive user experience, the system should analyse wardrobe photos, detect faces, and generate avatar-based outfit visualisations in 5-10 seconds.
• Portability: The solution should work seamlessly between mobile platforms (Android and iOS) and desktop development environments. The React Native framework ensures cross-platform functionality.. .

• Security: User data, including personal photos, clothing data, and facial textures, must be transmitted and stored securely. To protect user information, systems for authentication and encryption should be introduced.
• Maintainability: Modular system architecture allows for easy updates and replacement of specific components, such as AI models, AR visualisation modules, or database systems, without impacting the overall system.
• Usability: Users should be able to effortlessly rotate avatars, switch outfits, and preview clothing through responsive and interactive AR visualisations.

4.3 System Requirements

System requirements specify the hardware and software resources required to build and run the proposed AI-based smart wardrobe system with AR outfit visualisation. These criteria enable image processing, machine learning model execution, avatar rendering, and mobile app interaction.
The solution is intended to work on both mobile devices and desktop development environments, providing accessibility and flexibility.

4.4 Sample Use Case Scenarios

Scenario 01: User Creates Personalized 3D Avatar

Field	Description

Use case ID	UC-001
Use case name	Personalized 3D Avatar Creation
Primary actor	Mobile App User
Preconditions	• The user installs and logs into the mobile app. • The smartphone's camera access is enabled. • The user has a reliable internet connection to communicate with the backend server.
Main flow	• The user opens the mobile app and chooses the Create Avatar option. • The user shoots a selfie or uploads a facial photograph. • The image is transmitted to the backend server using the React Native → Node.js → Python processing pipeline. • The Python module recognises the user's face via OpenCV or MediaPipe. • The recognised face is trimmed and saved as a face texture image. • The system loads the underlying 3D avatar model developed with Blender. • The face texture is applied to the avatar's head material through Unity. • The user provides bodily dimensions such as height, waist, and shoulder size. • The avatar is scaled to match the measurements specified. • A personalised avatar is created and shown in the mobile app.
Extension	• If the uploaded image is unclear, the system prompts the user to take another selfie. • If face detection fails, the system will urge the user to alter the illumination or camera location.

Scenario 02: AR Outfit Visualization

Field	Description
Use case ID	UC-002
Use case name	AR-Based Outfit Visualization
Primary actor	Mobile App User
Preconditions	• The user has already made a personalised 3D avatar. • The smartphone offers augmented reality and camera access. • Outfit models are stored in the system wardrobe database.
Main flow	• The user selects the Outfit Visualisation function in the mobile app. • The system loads the user's personal 3D avatar. • The user chooses clothes items like shirts, trousers and shoes. • The specified apparel models are loaded as 3D assets (FBX). • The clothing models are connected to the avatar skeletons. • Mixamo animations are used for standing, walking, and turning. • The system activates the smartphone's camera for augmented reality visualisation. • The environment surface is detected using the Unity AR Foundation. • The avatar in the chosen attire is placed in the AR space. • The user can rotate, observe, and interact with the avatar in a real-world setting.

Extension	• If the device does not support AR, the system will display a 3D preview mode instead. • If a clothing model fails to load, the system prompts the user to select another outfit.

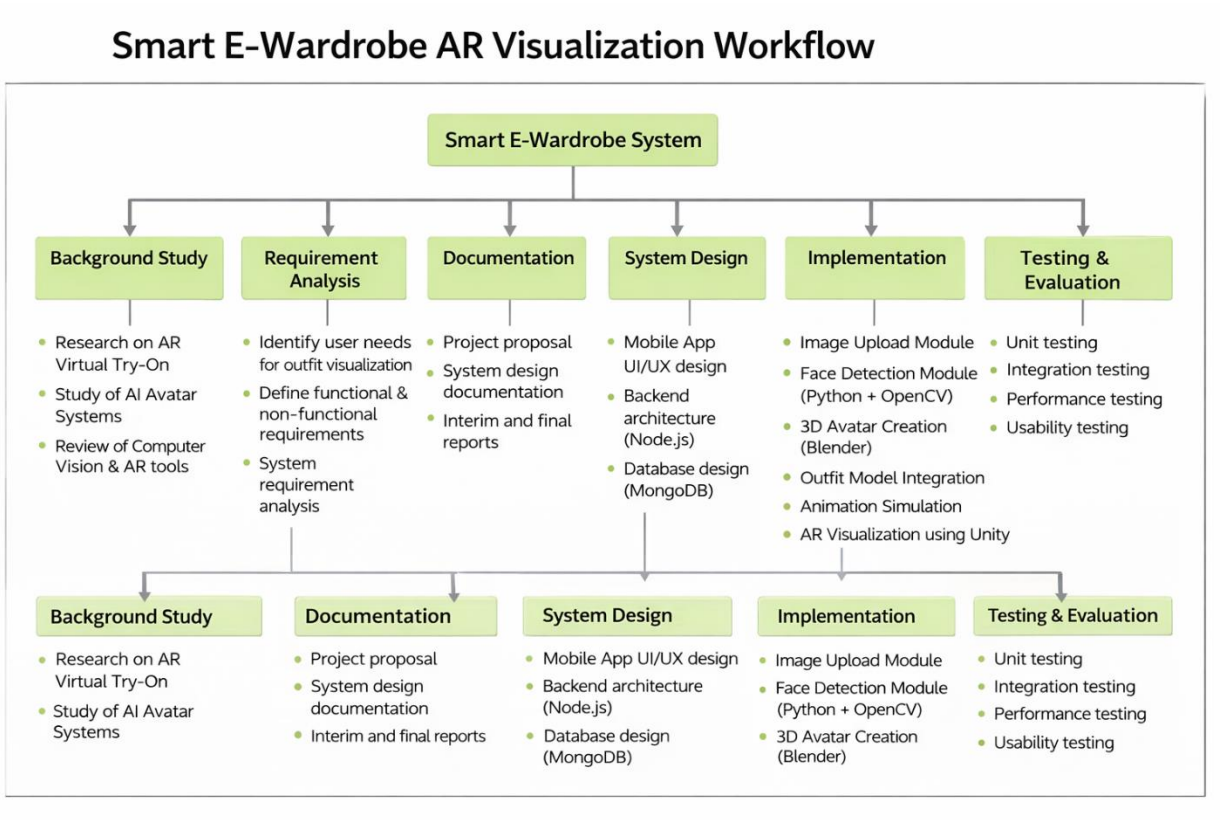


Figure 4.5 : Overall System Diagram



Figure 4.6 : system work flow

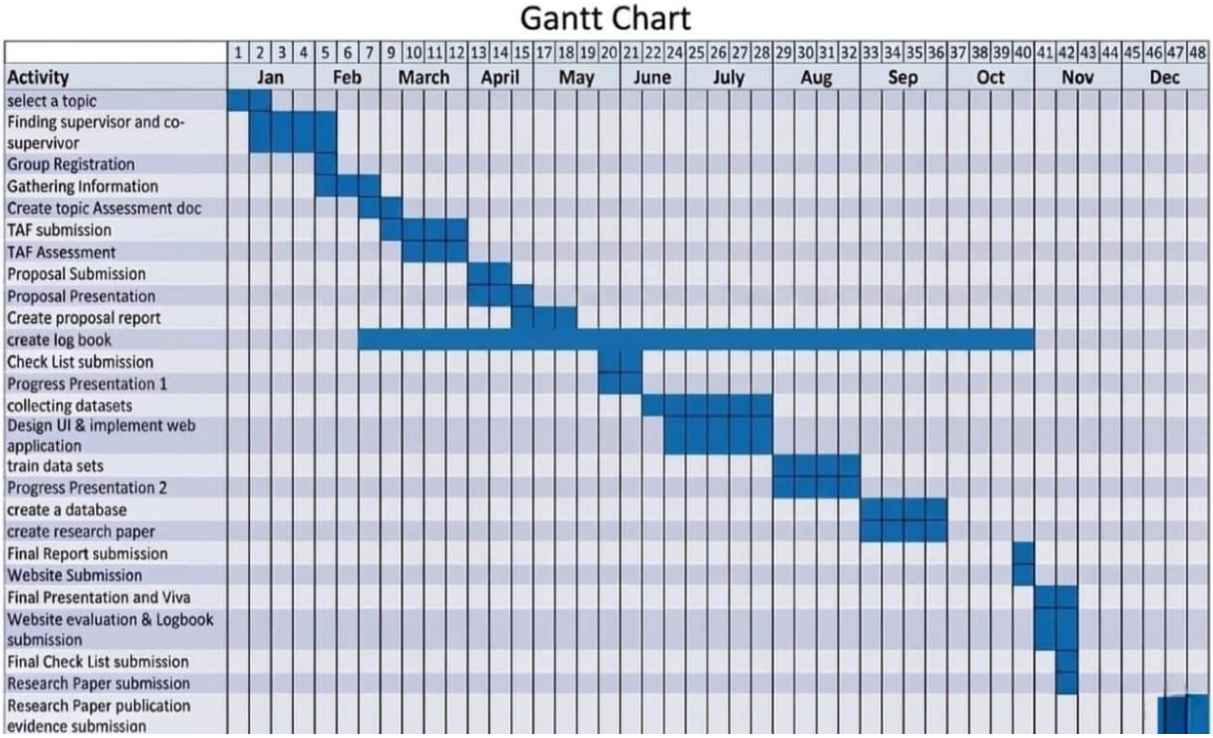


Figure 4.7 : Grant chart

5. DESCRIPTION OF PERSONNEL AND FACILITIES

5.1 Personnel

*This component of the research is carried out independently by **viduni vinethma** , Student ID **iT22170316** an undergraduate student in the **BSc. (Hons) in Information Technology** specializing in Information Technology at the **Sri Lanka Institute of Information Technology**.*

*The researcher is responsible for the design, implementation, and evaluation of the **AI-Based Outfit Visualization with Animated Fitting System**, which enables users to visualize outfits on a personalized 3D avatar using augmented reality.*

Researcher Responsibilities

• ***Face Detection and Processing***
Implement face detection algorithms using Python libraries such as OpenCV or MediaPipe to identify and extract facial features from user-uploaded images and generate a face texture for avatar rendering.

• ***3D Avatar Generation***
Design and create a customizable 3D human avatar using Blender, including body structure components such as head, body, arms, and legs. Export the avatar model in FBX format for integration with the visualization engine.

• ***Face Texture Mapping***
Develop methods to map the extracted facial texture onto the avatar head using Unity material and shader systems, allowing the avatar to visually resemble the user.

• ***Body Measurement Scaling***
Implement avatar scaling algorithms that adjust the avatar's body proportions based on user-input measurements such as height, waist, shoulder width, and weight to improve visualization accuracy.

• ***Clothing Model Integration***
Convert clothing items into optimized 3D models and attach them to the avatar skeleton structure to simulate real outfit fitting.

• ***Animation and Fitting Simulation***
Integrate animation sequences using Mixamo and Unity animation controllers to simulate natural body movements such as walking, standing, and turning to provide realistic outfit fitting visualization.

• ***Augmented Reality Visualization***
Develop AR visualization features using Unity AR Foundation to place the animated avatar in the real environment through the mobile camera, enabling users to preview outfits interactively.

• ***Mobile Application Integration***
Develop the mobile application interface using React Native (Expo) and integrate backend services using Node.js and Python for face processing and avatar generation.

• ***Evaluation and Testing***
Conduct usability testing and performance evaluation of the visualization system to assess accuracy, realism, and user satisfaction.

Supervision

The project is supervised by an academic supervisor from the Department of Information Technology who provides guidance on research methodology, system architecture, and experimental evaluation.

5.2 Facilities

Computing Resources

• Personal laptop with Intel Core i7 processor, 16GB RAM, AMD Ryzen 5000 Series processor and AMD Radeon Graphics for development and testing.

• Access to NVIDIA RTX 30 Series GPU with 6GB VRAM for running AI-based image processing and avatar rendering modules.

• Access to high-performance workstations available in the computing laboratories of the ***Sri Lanka Institute of Information Technology****.*

• Cloud computing platforms such as ***MongoDB*** *and* ***Google Colab*** *for model development, data processing, and experimental testing.*

Software and Tools

• ***Development Environment:***
Visual Studio Code, PyCharm

• ***Mobile Development:***
React Native with Expo

• ***Backend Development:***
Node.js and Express.js

• ***Programming Languages:***
Python, JavaScript

• ***Computer Vision Libraries:***
OpenCV, MediaPipe

• ***3D Modeling:***
Blender for avatar and clothing model creation

• ***AR and 3D Visualization:***
Unity with AR Foundation

• ***Animation Tools:***
Mixamo for generating avatar movement animations

• ***Database:***
MongoDB

6 . Design Constraints

During the design and implementation of the AI-based Smart E-Wardrobe System with AR Outfit Visualisation, numerous technical and practical obstacles arise.

• Complex integration of multiple technologies.
The system requires the integration of a number of technologies, including React Native mobile development, Node.js backend services, Python-based computer vision processing, Unity AR visualisation, and 3D modelling tools. Managing compatibility between various components might be a technological challenge.

• Limited availability of 3D clothing models.
Realistic clothing simulations require well-designed 3D garment models. Obtaining high-quality models or turning current apparel photos into precise 3D models may necessitate additional time and design resources.

Mobile Device Hardware Limitations
Not all smartphones can perform complex AR processing or real-time 3D rendering. Devices with limited processing power or memory may perform poorly while rendering animated avatars.

• Face Detection Accuracy: Poor lighting, occlusions, or unique angles can lead to inaccurate results for face detection algorithms, affecting the quality of the created avatar.

• Real-Time Rendering Performance Constraints: Creating animated avatars and clothing models in augmented reality needs significant processing resources. Optimising performance to ensure smooth visualisation on mobile devices can be difficult.

• Time and resource constraints.
Careful project planning is required while developing AI models, 3D avatars, AR visualisation modules, and mobile applications at the same time. Delays in model training, animation design, and AR integration may have an impact on the project's total timetable 3.6 Budget Justification

The requested funding will be used to build and test the AI-Based Smart E-Wardrobe System with AR Outfit Visualisation. To properly develop, implement, and assess the system, a number of software tools, cloud services, and hardware resources are

needed.

The software costs include development environments, AR development tools, and supporting libraries needed for mobile application development, machine learning model implementation, and 3D visualisation. Although certain tools, such as Python, Blender, and Unity, provide free versions, extra plugins, assets, and development utilities may demand licensing or a subscription charge.

Item	Estimated amount for a 6-month period (SL rupees)
Licensed Software Tools	10000.00
Cloud Services / Hosting	10000.00
Online Courses / Workshops	10000.00
External Storage	10000.00
Internet / Networking	12000.00
Other Hardware	10000.00
Total	62000.00

Table 3

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